KAIZEN MATHS

Lesson starter

Ratio and Proportion

Proportion

Name

S1

If 8 items cost £80, find the cost of 11 items.

S2

8 workers can complete a job in 12 days. How long will it take 12 workers?

S3

If 7 items cost £42, find the cost of 12 items.

S4

6 workers can complete a job in 9 days. How long will it take 9 workers?

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NUM-B9A2-L3 · S

Show your working

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Exit ticket

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If 3 items cost £15, find the cost of 14 items.

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4 workers can complete a job in 7 days. How long will it take 14 workers?

E3

A recipe for 3 people needs 36g flour. How much flour for 12 people?

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Teacher answers & working

Ratio and Proportion · Proportion

NUM-B9A2-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

If 8 items cost £80, find the cost of 11 items.

Answer:

£110

Working:

This is direct proportion.

Find the value for 1 item, 1 hour, or 1 person first.

Unit value = 80 ÷ 8 = 10

New value = 10 × 11 = 110

Shopping costs increase proportionally with quantity.

S2

8 workers can complete a job in 12 days. How long will it take 12 workers?

Answer:

8 days

Working:

This is inverse proportion: more workers means fewer days for the same job.

Total work units = 8 × 12 = 96

Keep the same total work units and divide by the new number of workers.

New time = 96 ÷ 12 = 8 days

For the same job, workers × days stays constant.

S3

If 7 items cost £42, find the cost of 12 items.

Answer:

£72

Working:

This is direct proportion.

Find the value for 1 item, 1 hour, or 1 person first.

Unit value = 42 ÷ 7 = 6

New value = 6 × 12 = 72

Shopping costs increase proportionally with quantity.

Teacher answers & working

Ratio and Proportion · Proportion

NUM-B9A2-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

6 workers can complete a job in 9 days. How long will it take 9 workers?

Answer:

6 days

Working:

This is inverse proportion: more workers means fewer days for the same job.

Total work units = 6 × 9 = 54

Keep the same total work units and divide by the new number of workers.

New time = 54 ÷ 9 = 6 days

For the same job, workers × days stays constant.

E1

If 3 items cost £15, find the cost of 14 items.

Answer:

£70

Working:

This is direct proportion.

Find the value for 1 item, 1 hour, or 1 person first.

Unit value = 15 ÷ 3 = 5

New value = 5 × 14 = 70

Shopping costs increase proportionally with quantity.

E2

4 workers can complete a job in 7 days. How long will it take 14 workers?

Answer:

2 days

Working:

This is inverse proportion: more workers means fewer days for the same job.

Total work units = 4 × 7 = 28

Keep the same total work units and divide by the new number of workers.

New time = 28 ÷ 14 = 2 days

For the same job, workers × days stays constant.

Teacher answers & working

Ratio and Proportion · Proportion

NUM-B9A2-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

A recipe for 3 people needs 36g flour. How much flour for 12 people?

Answer:

144g

Working:

This is direct proportion.

Find the value for 1 item, 1 hour, or 1 person first.

Unit value = 36 ÷ 3 = 12

New value = 12 × 12 = 144

Recipe ingredients scale directly with servings.